

Lab3 - VLAN Configuration

1. Network Topology & Architecture

The network consists of three switches (**S1**, **S2**, **S3**) interconnected via trunk links, with end devices spread across different floors or locations but logically segmented into Virtual Local Area Networks (VLANs).

- **S1 (Core/Distribution Switch):** Acts as the central aggregation point connecting to S2 and S3.
- **S2 (Access Switch):** Hosts PC1, PC2, and PC3.

S3 (Access Switch): Hosts PC4, PC5, PC6, and a Cisco IP Phone on \$F0/11\$.

2. Complete Device Configurations

Switch 1 (S1) Configuration

S1 requires the creation of all global VLAN databases. Trunking commands are implied on its uplink interfaces to pass multi-VLAN traffic to S2 and S3.

```
S1> enable
S1# configure terminal

! Configure VLAN Names
S1(config)# vlan 10
S1(config-vlan)# name Faculty/Staff
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name Students
S1(config-vlan)# exit

S1(config)# vlan 30
S1(config-vlan)# name Guest(Default)
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name Management&Native
S1(config-vlan)# exit

S1(config)# vlan 150
```

```
S1(config-vlan)# name VOICE
S1(config-vlan)# exit
```

Switch 2 (S2) Configuration

S2 handles local access ports for PC1, PC2, and PC3.

```
S2> enable
S2# configure terminal

! Configure Global VLANs
S2(config)# vlan 10
S2(config-vlan)# name Faculty/Staff
S2(config-vlan)# vlan 20
S2(config-vlan)# name Students
S2(config-vlan)# vlan 30
S2(config-vlan)# name Guest(Default)
S2(config-vlan)# vlan 99
S2(config-vlan)# name Management&Native
S2(config-vlan)# vlan 150
S2(config-vlan)# name VOICE
S2(config-vlan)# exit

! Assign Access Ports
S2(config)# interface FastEthernet0/11
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
S2(config-if)# exit

S2(config)# interface FastEthernet0/18
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 20
S2(config-if)# exit

S2(config)# interface FastEthernet0/6
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 30
S2(config-if)# exit
```

Switch 3 (S3) Configuration

S3 requires access port assignments and unique Quality of Service (QoS) and Voice VLAN mapping on interface \$F0/11\$.

```
S3> enable
S3# configure terminal

! Configure Global VLANs
S3(config)# vlan 10
```

```

S3(config-vlan)# name Faculty/Staff
S3(config-vlan)# vlan 20
S3(config-vlan)# name Students
S3(config-vlan)# vlan 30
S3(config-vlan)# name Guest(Default)
S3(config-vlan)# vlan 99
S3(config-vlan)# name Management&Native
S3(config-vlan)# vlan 150
S3(config-vlan)# name VOICE
S3(config-vlan)# exit

```

```

! Assign Access Ports for Data
S3(config)# interface FastEthernet0/18
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 20
S3(config-if)# exit

```

```

S3(config)# interface FastEthernet0/6
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 30
S3(config-if)# exit

```

```

! Assign Voice + Data and Enable QoS on F0/11
S3(config)# interface FastEthernet0/11
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 10
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
S3(config-if)# exit

```

3. Master Addressing Table

Device	Interface	IP Address	Subnet Mask	Assigned VLAN
PC1	NIC	172.17.10.21	255.255.255.0	VLAN 10 (Faculty/Staff)
PC2	NIC	172.17.20.22	255.255.255.0	VLAN 20 (Students)
PC3	NIC	172.17.30.23	255.255.255.0	VLAN 30 (Guest)

Device	Interface	IP Address	Subnet Mask	Assigned VLAN
PC4	NIC	172.17.10.24	255.255.255.0	VLAN 10 (Faculty/Staff)
PC5	NIC	172.17.20.25	255.255.255.0	VLAN 20 (Students)
PC6	NIC	172.17.30.26	255.255.255.0	VLAN 30 (Guest)

4. Verification & Testing Playbook

Step A: Verify Switch VLAN Maps

To quickly check that your names and interface associations match the lab design:

S2# show vlan brief

S2#show vlan brief			
VLAN Name	Status	Ports	
-----	-----		
1 default	active	Fa0/2, Fa0/3, Fa0/4,	
Fa0/5		Fa0/7, Fa0/8, Fa0/9,	
Fa0/10		Fa0/12, Fa0/13, Fa0/14,	
Fa0/15		Fa0/16, Fa0/17, Fa0/19,	
Fa0/20		Fa0/21, Fa0/22, Fa0/23,	
Fa0/24		Gig0/1, Gig0/2	
10 Faculty/Staff	active	Fa0/1, Fa0/11	
20 Students	active	Fa0/18	
30 Guest (Default)	active	Fa0/6	
99 Management&Native	active		
150 VOICE	active		
1002 fddi-default	active		
1003 token-ring-default	active		
1004 fddinet-default	active		
1005 trnet-default	active		

Step B: Host Network Layer Verification

On **PC1** Command Prompt:

```
C:\> ipconfig
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:21FF:FEDD:EA4B
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.17.10.21
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

Expected Output: IP Address.: 172.17.10.21 Subnet Mask
.: 255.255.255.0

Step C: Inter-VLAN / Connectivity Testing

If you attempt to ping **PC4** from **PC1** immediately after this configuration:

```
C:\> ping 172.17.10.24
```

```
C:\>ping 172.17.10.24

Pinging 172.17.10.24 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.17.10.24:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Result: Request timed out.

```
C:\> tracert 172.17.10.24
```

```
C:\>tracert 172.17.10.24

Tracing route to 172.17.10.24 over a maximum of 30 hops:

 1  *          *          *          Request timed out.
```

Result: Drops immediately at the local node.

5. Critical Analysis & Troubleshooting

Why did the pings fail?

Although PC1 and PC4 are structurally assigned to the correct logical broadcast domain (VLAN 10), the switches are linked together via standard links operating inside **VLAN 1** by default.

- Looking at the provided `show vlan brief` output for S2, ports **Gig0/1** and **Gig0/2** are still explicitly stuck in **VLAN 1**.
- Because these inter-switch links are operating as access ports in VLAN 1, any frame tagged with **VLAN 10**, **VLAN 20**, or **VLAN 30** traveling from S2 trying to cross over to S1 will be completely dropped.

How to resolve this issue?

To allow traffic from all configured VLANs to travel between S2, S1, and S3, the interconnected ports (such as \$Gig0/1\$ and \$Gig0/2\$) must be changed from Access Mode to **Trunk Mode** using IEEE 802.1Q encapsulation.

Remediation Command Example (On S1, S2, and S3 uplink ports):

```
S2(config)# interface range g0/1 - 2
S2(config-if-range)# switchport mode trunk
S2(config-if-range)# switchport trunk native vlan 99
```

```
S2(config)#interface range g0/1 - 2
S2(config-if-range)#switchport mode trunk
```

```
S2(config-if-range)#switch trunk native vlan 99
```

```
S2#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,99,150

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20,30,150
```

6. Personal Reflections

- **Logical vs. Physical Architecture:** This lab highlights that physical proximity or simply being on the same IP subnet is no longer enough to guarantee data delivery once layer 2 micro-segmentation is introduced.

The Magic of Voice Integration: Merging voice traffic (VLAN 150) and data traffic (VLAN 10) down a single copper drop on S3 (F0/11) shows how enterprise networks maximize their cabling investments. Using QoS (mls qos trust cos) prevents an intensive PC download from degrading a crisp VoIP voice call.